

PUBLISHED

Question	Answer	Marks	Guidance
3	<i>independent variable</i> 1 at least three different light intensities ; <i>dependent variable</i> 2 the colour of the indicator ; 3 and 4 details of the method ;; - method of changing light intensity e.g. different lamp distances / dimmer switch altered / different bulbs / different lamps / different power lamps / detail of how to make one in the dark and one in the light - method to avoid heating effect of light e.g. heat shield / beaker of water between lamp and plant / use LED bulbs to prevent heating - keeping the plant under the water / weighting the plant down - sample the water around the plant and test with indicator 5 and 6 and 7 variables kept constant ;;; - colour / wavelength, of light - temperature - type / species / age, of plant - size / mass / length / height / surface area / number of leaves, of plant - volume of water (covering plant) - volume / number of drops, of (hydrogencarbonate) indicator - concentration of (hydrogencarbonate) indicator - set (stated) time for equilibration (to the new light intensity) - set (stated) time before noting colour 8 two or more repeats ; 9 relevant safety ;	6	